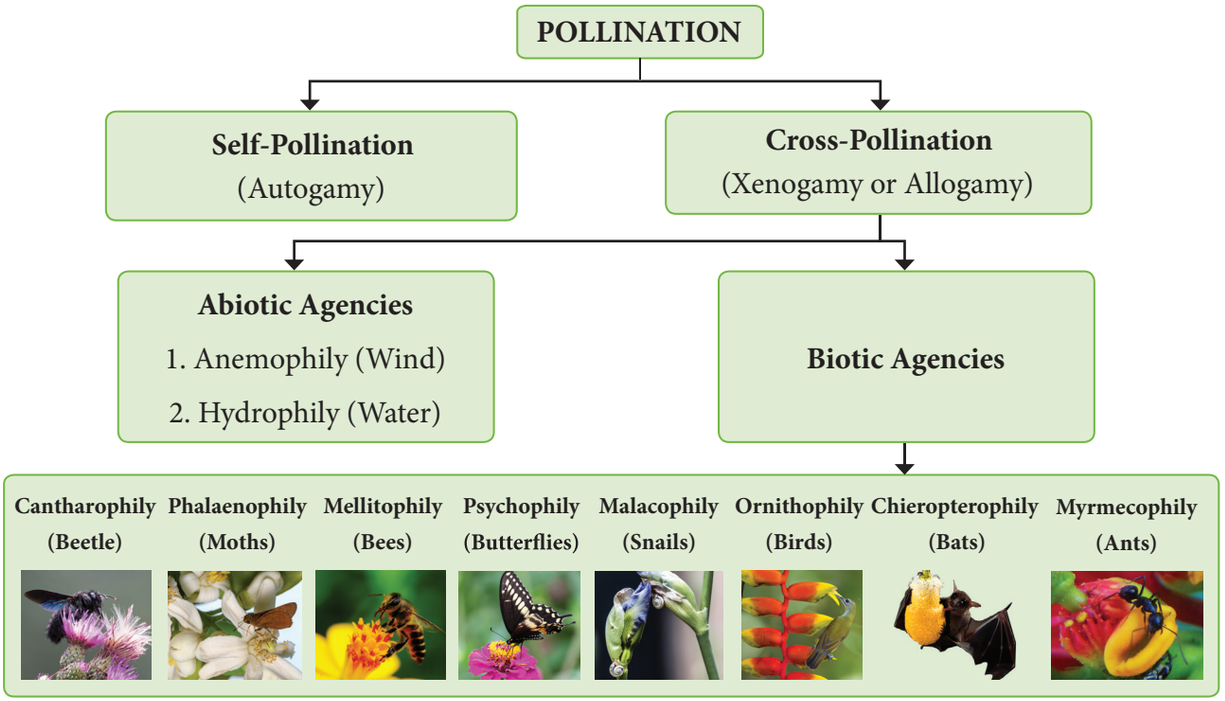


is same as that of sea water, so that, the pollen floats freely at any depth. The stigma is very large and long. The pollen comes in contact with the stigma and gets coiled around the stigma thus effecting pollination.

Biotic agent

Zoophily: Pollination by the agency of animals is called zoophily and flowers are said to be zoophilous. Animals that bring about pollination may be birds, bats, snails and insects. Of these, insects are well adapted to bring pollination. Larger animals like primates (lemurs), arboreal rodents, reptiles (gecko lizard and garden lizard) have also been reported as pollinators.

a. Ornithophily: Pollination by birds is called Ornithophily. Some common plants that are pollinated by birds are *Erythrina, Bombax, Syzygium, Bignonia, Sterlitzia* etc., Humming birds, sun birds, and honey eaters are some of the birds which regularly visit flowers and bring about pollination.

The ornithophilous flowers have the following characteristic features:

- The flowers are usually large in size.
- The flowers are tubular, cup shaped or urn-shaped.
- The flowers are brightly coloured, red, scarlet, pink, orange, blue and yellow which attracts the birds.
- The flowers are scentless and produce nectar in large quantities. Pollen and nectar form the floral rewards for the birds visiting the flowers.
- The floral parts are tough and leathery to withstand the powerful impact of the visitors.

b. Cheiropterophily: Pollination carried out by bats is called cheiropterophily. Some of the common cheiropterophilous plants are *Kigelia africana*, *Adansonia digitata*, etc., Bats are nocturnal and are attracted by the odour of the flowers that open at or after dusk. The cheiropterophilous plants have flowers borne singly or in clusters quite away from the leaves and branches either on the long peduncle or on the trunk or branches. The flowers produce large quantities of nectar.

Pollination in *Adansonia digitata*: In this plant, the ball of stamens and the stigma project beyond the floral envelops. The bat holds the flower by clasping the stamen ball to its breast. While taking nectar its breast becomes laden with numerous pollen grains,

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